

Capital Efficiency in the Post-Digital Film Industry:

An End-to-End Machine Learning Framework for ROI Prediction and Risk Assessment

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Abstract

The global motion picture industry is currently navigating a period of profound structural disruption, characterized by a phenomenon we term the “Revenue Paradox.” While aggregate Gross Box Office (GBO) receipts have returned to pre-pandemic levels, underlying Net Profit Margins have declined due to a persistent “Cost Squeeze” driven by escalating production budgets and diminishing marketing returns. Traditional investment heuristics relying on vanity metrics and artistic intuition fail under these non-linear dynamics.

This study bridges financial econometrics and computer science through a dual-layer investment framework. Using descriptive analytics on 5,844 commercial films (2000–2025), we identify systemic market failures, including a 27% collapse in marketing efficiency and a persistent “Capital Death Valley” for mid-budget productions. These insights are operationalized through a Stacking Ensemble Machine Learning model. By introducing “Time-Travel” rolling aggregation for star power and inflation-adjusted normalization, the model achieves an R^2 of **63.74%** and an RMSE of 1.1002 on a log-transformed scale. The findings support a data-driven “Barbell Investment Strategy,” reallocating capital from mid-tier dramas toward low-budget horror (cash-flow stability) and high-budget franchise IPs (growth engines).

Keywords: Capital Efficiency, Film Economics, ROI Prediction, Stacking Ensemble, Risk Assessment, Strategic Portfolio Management.

1 Introduction

The motion picture industry has historically been a high-risk investment domain governed by extreme revenue skewness, where a small number of blockbusters subsidize systemic losses. In the post-digital era (2020–present), these risks have intensified as cost structures inflate faster than revenue recovery.

1.1 The Economic Problem: Profitless Growth

Despite the resurgence of theatrical attendance during 2022–2025, profitability remains constrained. Vogel [1] observes that rising production and marketing expenditures have eroded margins, creating a “Revenue Paradox” in which studios report higher gross receipts but weaker free cash flow. Consequently, the investment challenge has shifted

from hit prediction to capital efficiency optimization.

1.2 The Technical Problem: Data Leakage

Many predictive studies suffer from look-ahead bias, computing features such as actor popularity using post-release information. This violates temporal causality and invalidates real-world deployment. Addressing this flaw is critical for actionable decision support.

2 Related Work

Early predictive studies employed neural networks and decision trees for revenue classification but lacked economic normalization and temporal awareness. De Vany and Walls [2] demonstrated that film revenues follow a Levy-stable distribu-

tion, implying inherent volatility unaffected by star participation. More recent ML-based approaches often rely on static datasets (e.g., TMDB 5000), inadvertently introducing data leakage. Our work addresses these gaps through inflation-adjusted normalization and strictly causal feature construction.

3 System Implementation

The proposed system integrates three layers: data ingestion, analytical visualization, and predictive modeling.

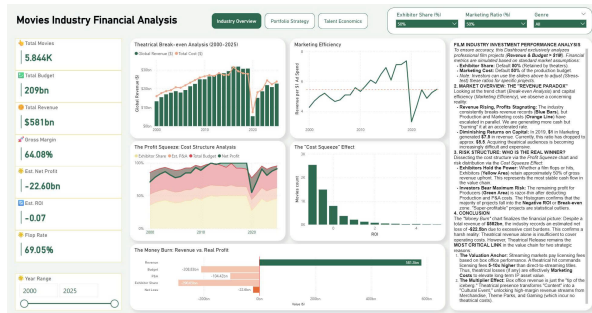


Figure 1: System architecture integrating ETL, descriptive analytics, and predictive modeling.

4 Descriptive Market Analytics

An exploratory analysis was conducted on 5,844 commercial films with budgets and revenues exceeding \$1M.

4.1 The Profit Squeeze

As illustrated in Fig. 2, revenue growth is decoupled from profit.

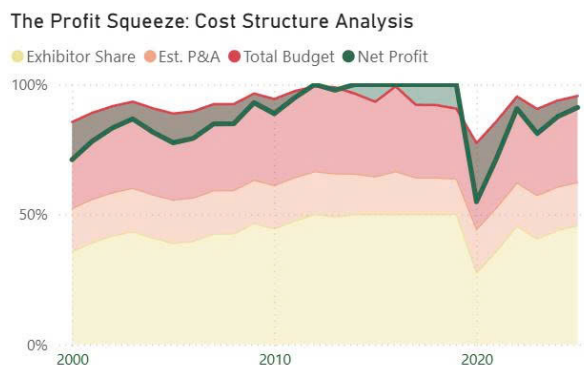


Figure 2: Divergence between global revenue growth and total cost escalation (The Profit Squeeze).

4.2 Marketing Efficiency Collapse

Marketing ROI has declined significantly post-pandemic.



Figure 3: Decline in marketing multiplier indicating rising acquisition costs.

4.3 Budget Polarization

The market exhibits a "missing middle" structure.

Budget Tier Matrix

Win Rate % by Budget Segments

Genre	< \$10M	\$10M - \$50M	\$50M - \$100M	> \$100M
Action	Green	Green	Green	Green
Adventure	Green	Green	Green	Green
Animation	Green	Green	Green	Green
Comedy	Green	Green	Green	Green
Crime	Green	Green	Green	Green
Drama	Green	Green	Green	Green
Family	Green	Green	Green	Green
Fantasy	Green	Green	Green	Green
History	Green	Green	Green	Green
Horror	Green	Green	Green	Green
Music	Green	Green	Green	Green
Mystery	Green	Green	Green	Green
Romance	Green	Green	Green	Green
Science Fiction	Green	Green	Green	Green
Thriller	Green	Green	Green	Green
War	Green	Green	Green	Green
Western	Green	Green	Green	Green

Figure 4: The "Budget Death Valley": Mid-budget films (\$20M–\$50M) exhibit persistent underperformance.

5 Methodology

5.1 Inflation Adjustment

All financial values were normalized to 2024 USD to ensure comparability:

$$R_{adj} = R_t \times \frac{CPI_{2024}}{CPI_t} \quad (1)$$

5.2 Rolling Star Power

We calculate star power using strictly past data:

$$S_{a,t} = \frac{1}{k} \sum_{i=1}^k \ln(r_{t-i} + 1) \quad (2)$$

5.3 Stacking Ensemble

The predictive architecture combines XGBoost, LightGBM, and Gradient Boosting models with a linear meta-learner to capture non-linear market dynamics.

6 Experimental Results

6.1 Model Performance

The Stacking Ensemble outperforms single models across all metrics.

Table 1: Model Performance (Log Scale)

Model	RMSE	MAE	R^2
Linear Reg.	1.1438	0.8864	0.6082
Random Forest	1.1047	0.8561	0.6345
XGBoost	1.1021	0.8519	0.6362
Stacking	1.1002	0.8439	0.6374

6.2 Feature Importance

Feature analysis confirms that Budget and Franchise status are the dominant predictors of revenue.

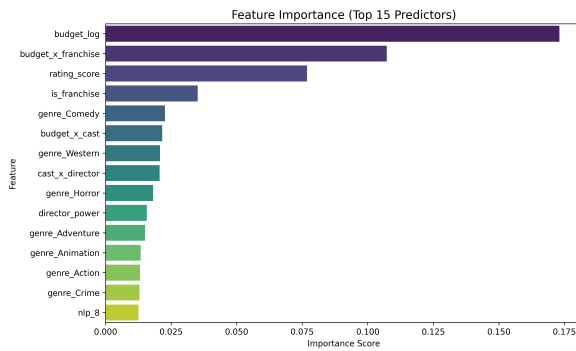


Figure 5: Top 15 Feature Importance.

7 Strategic Implications

The results support a **Barbell Investment Strategy**:

- Defensive:** Allocate 40% capital to low-budget (<\$10M) Horror/Thriller films.
- Offensive:** Allocate 40% to high-budget (>\$100M) Franchise IPs.
- Avoid:** The mid-budget (\$20M–\$50M) range, identified as the "Death Valley" of capital efficiency.

8 Limitations Conclusion

This study focuses primarily on box office revenue and does not fully capture downstream streaming income. However, by integrating macro-level analytics with micro-level prediction, this framework provides a robust tool for capital-efficient decision-making in the modern film industry.

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